Yohan Senadheera

Site Reliability Engineering Intern | Computer Engineering Undergraduate

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SUMMARY

Site Reliability Engineering Intern at WSO2 and final-year Computer Engineering undergraduate working on DevOps, CI/CD, cloud infrastructure and security for multi-cloud systems. Designed a pull-based CI/CD control plane for AWS, Azure and on-premises data planes, and automated CA-signed certificate issuance and rotation across 20+ microservices, replacing a fully manual process. Author of Terraform modules merged into WSO2's public Azure module library and contributor to upstream Argo Workflows. Strong in Kubernetes, Terraform, Helm, GitOps, Linux troubleshooting and observability (Prometheus, Grafana, Alertmanager), with security fundamentals in IAM, secrets and certificates. Mentored 40+ students as a university instructor, edited for a 200+ member technical community, and captained a university team to an inter-university championship title. Communicates early, learns new tools quickly and takes ambiguous problems from design to documented, maintainable delivery.

TECHNICAL SKILLS

CI/CD & Automation: Azure DevOps (YAML pipelines), Argo Workflows, Argo Events, Argo CD, GitHub Actions, GitOps, Git
IaC & Configuration: Terraform (reusable modules), Helm, declarative configuration management with GitOps (Argo CD)
Containers & Orchestration: Docker, Kubernetes (deployments, services, scheduling, RBAC), AKS, EKS, k3s
Cloud: AWS (EC2, S3, IAM, EKS, IRSA), Azure (AKS, Entra ID, Key Vault, NSGs, Workload Identity Federation)
Observability & Reliability: Prometheus, Grafana, Alertmanager, alerting rules, HA design, chaos testing, runbooks
Linux & Scripting: Linux troubleshooting, Bash, Python, OpenSSL, Azure CLI, SQL
Security & Networking: IAM, secrets and certificate management, OIDC, mTLS, PKI, least-privilege RBAC, TCP/IP, routing

EXPERIENCE

Site Reliability Engineering Intern — **Apr 2026 – Present**
WSO2 (Pvt) Ltd — *Colombo, Sri Lanka*

Multi-Cloud CI/CD Platform

- Designed a pull-based, cloud-agnostic control plane for AWS, Azure and on-premises data planes, with zero inbound connections into production: every data plane dials out to the control plane.
- Built the pipeline layer on Argo Workflows, Argo CD (Git-synced clusters) and NATS pub/sub task delivery
- Replaced shared admin credentials with network isolation and RBAC restrictions and removed static secrets from the platform using an mTLS message bus with per-identity ACLs and an OIDC issuer.
- Built high availability and recovery into the provisioning: multi-AZ clusters, replicated workflow controllers with leader election, and a 3-node NATS JetStream cluster with durable task streams that redeliver missed work.
- Authored Terraform modules that provision the complete Argo setup, submitted to WSO2's public AWS and Azure module libraries (aws-terraform-modules, azure-terraform-modules).
- Contributed to the upstream Argo Workflows public repository for gaps found in this work (Argo Workflows).
- Diagnosed and fixed reliability issues from live logs and resource monitoring, including VM memory pressure, pods scheduled onto the wrong AKS node pool, a secret-sync race condition and a silently failing credential refresh.
- Replaced 50+ inline sed-based secret injections in the existing pipeline templates with External Secrets Operator, syncing from Azure Key Vault and AWS Secrets Manager through workload identity.

Tech: Argo Workflows, Argo Events, Argo CD, NATS JetStream, Kubernetes, AKS, EKS, k3s, Helm, Terraform, OIDC, AWS, Azure

PKI Certificate Automation

- Replaced a fully manual, per-service self-signing process with automated CA-signed certificate issuance and rotation across 20+ microservices, delivering the migration as pull requests across 7 repositories.
- Built the pipeline automation in Bash and Azure CLI: it triggers the central CA pipeline across organisations, polls it to completion, and uploads the signed certificate to the key vault.
- Designed the security model: cross-organisation trust via Workload Identity Federation replaced personal access tokens, with least-privilege identity segmentation so no single compromised credential can run the signing workflow.
- Contributed the project's authentication and authorisation Terraform modules to WSO2's public Azure module library through the formal Change Request process.

Tech: Terraform, Azure DevOps YAML, Bash, OpenSSL, Azure Key Vault, Workload Identity Federation, Azure RBAC, GitHub

Casual Instructor – Database Systems & Programming Methodology — **Nov 2024 – Aug 2025**
Department of Computer Engineering, University of Peradeniya — *Peradeniya, Sri Lanka*

- Taught and mentored 40+ students in database systems and C programming, reviewing code and designs in labs and one-to-one sessions and giving direct, constructive feedback before implementation.

Trainee – People's Bank (branch banking operations, customer records and data entry) May 2021 – Feb 2022

PROJECTS

Cloud-Native Platform with Kubernetes Observability & Recovery | GitHub Feb 2026
Built a Kubernetes platform running a containerised FastAPI service and PostgreSQL database with Prometheus metrics, a Grafana golden-signals dashboard, Alertmanager alerting and readiness-gated self-healing, validated with chaos experiments and documented in runbooks.
Tech: Kubernetes, Docker, Prometheus Operator, Grafana, Alertmanager, PostgreSQL, Python, Linux

Private 5G Network Implementation – with SLT Mobitel Aug 2025 – Present
Deployed the Aether SD-Core 5G core on Kubernetes with Helm on a university server, and implemented an edge UPF on Raspberry Pi with a real 5G RAN to optimise user-plane traffic; teammates handled RAN virtualisation and core tuning.
Tech: Kubernetes, SDN, Wireshark, Helm, Aether SD-Core, Raspberry Pi

P-E-BO Desk Companion – Third Year Project | GitHub Dec 2024 – Jul 2025
Developed software for an AI desk companion robot that detects 7 emotions at 95% accuracy: a React Native (Expo) app with Firebase authentication and real-time device sync, AWS Rekognition face recognition backed by S3, Gemini-powered voice interaction, QR-code Wi-Fi provisioning and over-the-air updates.
Awards: IESL Undergraduate Inventor of the Year – Top 10 Finalist; ICIIS 2025 – 3rd Place.
Tech: React Native, Expo, Firebase, AWS , Gemini API, Raspberry Pi, Python

EDUCATION

BSc Eng (Hons), specialising in Computer Engineering **Feb 2022 – Oct 2026**
University of Peradeniya *Peradeniya, Sri Lanka*

- Relevant coursework: Advanced Communication Networks (networking, SDN, Docker), Network Security, Databases, Software Architecture.

GCE Advanced Level – Nalanda College, Colombo (4 A passes) 2020
GCE Ordinary Level – Nalanda College, Colombo (9 A passes, School Best Result) 2017

CERTIFICATIONS

Certified Kubernetes Administrator (CKA) – Udemy Expected Oct 2026
WSO2 Certified: Identity Server Practitioner – WSO2 Aug 2026
CCNA 1, 2 and 3 – Cisco Networking Academy (routing & switching, enterprise networking, security) Aug 2026
Docker / Kubernetes for Beginners – KodeKloud Mar 2026
AWS Technical Essentials – AWS Jan 2026
5G for Everyone – Qualcomm Sep 2025
Linux Commands & Shell Scripting – IBM May 2025

AWARDS & COMPETITIONS

IESL Undergraduate Inventor of the Year – Top 10 Finalist (P-E-BO) 2025
ICIIS 2025 – 3rd Place (P-E-BO) 2025
Coders V11.0 – 9th Place (ACES, University of Peradeniya) Oct 2024
MoraExtreme 9.0 – 27th Place (IEEE, University of Moratuwa) Oct 2024
Inter-University Air Rifle Championship – Men's Team Champions; Magam Shooting Sports Colours Aug 2024

LEADERSHIP & COMMUNITY

Editor – Hackers' Club, University of Peradeniya Aug 2025 – Jul 2026
Managed technical content and communications for a 200+ member community and coordinated committee events.

Team Captain – Shooting Sports Association, University of Peradeniya Jul 2023 – Oct 2025
Founded air gun shooting as a university sport, built a team drawn from all faculties, captained it for 2 years, and led it to the 2024 Inter-University Air Rifle Championship title.

Finance Vice Chair – Career Compass, IEEE Young Professionals Sri Lanka Apr 2025 – Sep 2025
Brand Manager / OC Vice President / Team Leader – AIESEC in Kandy Feb 2022 – Nov 2024
Senior Member – University Explorers' Club, University of Peradeniya Jul 2024 – Present

REFERENCES

Dr. Isuru Nawinne
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